

SMART INDIA HACKATHON 2026

VoxGuard

AI-Powered Real-Time Detection & Prevention of
Voice-Cloning Impersonation Attacks

COMPLETE PROJECT SPECIFICATION

Zero-to-Prototype Build & Product Handoff Document

Problem Statement ID	26104
Theme	Blockchain & Cybersecurity
Sponsor	AICTE
Category	Software
Document Purpose	Single-source technical + product reference for any implementation team to build a working prototype

This document consolidates the full research, architecture, product design, privacy design, threat modeling, and go-to-market reasoning developed for this project. It assumes no prior context — a reader or implementation team starting from zero should be able to build a working prototype from this document alone.

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1. Executive Summary

Modern AI voice-cloning tools can convincingly copy a person's voice from just a few seconds of audio — a voicemail, a YouTube clip, a video call. Criminals are using this to impersonate CEOs, officials, and family members on live phone calls to authorize fraudulent transfers or extract confidential information. Traditional defenses — caller ID, “I recognize their voice” — no longer work, because the human ear cannot reliably distinguish a well-made clone from the real person.

VoxGuard is a real-time voice-authenticity layer that listens to a call, analyzes the audio for spectral and prosody artifacts that give away synthetic speech, and delivers a risk score before the listener acts on what they just heard. It is built specifically to close a documented weak spot in existing anti-spoofing technology: state-of-the-art detectors score well in lab conditions but degrade sharply on real-world audio, and none of the current commercial players publish how well their systems work on Indian languages and accents specifically.

One-sentence USP: “We give any phone call a real-time ‘is this voice real’ risk score — built for the citizens, smaller institutions, and public-sector deployments that existing funded detectors don’t chase first — catching AI-cloned impersonation fraud in the moment it happens, not after the money is already gone.”

What this document contains

This is a single-source reference intended for a reader or implementation team with **zero prior context**. It covers the problem, the complete technical architecture, the exact AI model and why it was chosen, the research contribution unique to this project, the product’s real-world form factor, privacy and legal compliance design, failure-mode and trust handling, threat modeling against realistic attacks, cost justification, and a phased build roadmap — in enough detail that a development team could pick this up and build a working prototype without needing any other source material.

Project identity

Field	Value
Working project name	VoxGuard (final naming decision still open — see § 19)
Problem Statement	SIH26104 — AI-Powered Real-Time Detection & Prevention of Voice-Cloning Impersonation Attacks
Theme	Blockchain & Cybersecurity
Sponsor	AICTE
Core technique	Audio anti-spoofing via deep learning (AASIST), fine-tuned for Indian speech
Product form	Web app (P0) + Live-call mode (P1) + Bank/Telecom API-SDK (P2 roadmap)

2. The Problem — Explained From Zero

2.1 What voice cloning actually is

Voice cloning uses a type of AI model called a **text-to-speech (TTS)** or **voice conversion** system. Given a short sample of someone's real voice — sometimes as little as 5–10 seconds — the model learns that person's pitch, tone, rhythm, and speaking style well enough to generate entirely new sentences in their voice, saying things they never actually said. Free and low-cost tools that do this (e.g. Coqui TTS, RVC — Retrieval-based Voice Conversion) are now widely available, which is why this threat has grown so quickly.

2.2 Why humans can't catch it

Older cloning tools produced robotic, obviously fake audio. Current-generation models reproduce breathing patterns, emotional inflection, and natural pauses closely enough that even people who know the victim well cannot reliably tell the difference over a phone call, especially under the emotional pressure scammers deliberately create (urgency, a fabricated emergency, a demand for secrecy).

2.3 The scale of the problem in India

Rs. 70,000 Cr	550%	75% / 38%
Projected deepfake fraud losses in India in 2025 (Pi-Labs research)	Rise in deepfake-related cases in India since 2019 (Pi-Labs research)	Indians who've encountered deepfake content / who fell victim to a scam (McAfee survey)

These are not abstract numbers — real, documented cases include a company that lost **Rs. 10.70 crore** in a single deepfake voice-impersonation fraud in 2026, and an individual in Indore who lost **Rs. 1.83 lakh** after receiving a call cloned to sound like a relative.

2.4 Why existing defenses fail

- **Caller ID cannot be trusted.** It is a network-supplied label, not a cryptographic guarantee — legacy telecom signaling (SS7) allows the displayed number to be spoofed to show any number, including one already saved in the victim's contacts.
- **“I recognize their voice” no longer works** — that is precisely the assumption modern cloning defeats.
- **Manual callback verification helps, but isn't automatic** — most people don't think to do it under the pressure a scammer deliberately creates.

3. Who This Is For

Category	Who	Why they need this
Primary users	Individual citizens receiving personal calls; bank call-center agents; enterprise employees	They are the ones who directly hear the voice and make the decision to act
Secondary stakeholders	The impersonated individual; the institution bearing the financial loss	They suffer the consequence even though they're not on the call
Buyers / deployers	Banks, NBFCs, enterprises, telecom operators, public-sector/educational institutions	They want a reusable security layer for voice channels, as the PS explicitly frames it

4. Competitive Landscape — Honest Assessment

Three funded Indian companies already operate in adjacent space. This section is intentionally direct about what is and isn't genuinely differentiated — a judge, investor, or engineer will ask this, and an overclaimed answer is worse than an honest one.

Company	What they actually do (public info)	Overlap with VoxGuard
DetectifAI (2025, San Francisco / Bengaluru)	Audio-specific deepfake detection with live-call alerts. Founded by Tarini Padmanabhuni; 5 patents pending.	High — this is very close to VoxGuard's core “real-time, in-call” proposition.
Neural Defend (2024, San Francisco / Delhi)	Multimodal (image/video/audio) deepfake detection, API-as-a-service, proprietary/patented model. Citizen-facing upload-and-verify tool via a Zee News partnership.	Medium — they already have a citizen-facing tool, though not live-call, real-time.
Kroop AI (2021, Gandhinagar)	Audio/video/image deepfake detection for banking/finance client verification. Notably also builds synthetic voice/avatar generation tools.	Medium — enterprise-focused; dual-use (builds and detects) raises a trust question worth noting.

4.1 Important honesty note

Public marketing and funding coverage is not the same as verified technical documentation. This document does not claim certainty about these companies' internal model architecture, training data, or whether they publish per-language accuracy — those claims should be re-verified against primary sources before being stated as fact in front of a technical judge or investor.

4.2 The differentiator that actually holds up

“Real-time” and “serves citizens” are no longer uniquely VoxGuard's claims — competitors are moving into both. The genuinely defensible differentiator is the **quantified research contribution**: a documented, specific accuracy gap (see § 7.3) between lab performance and real-world performance, closed specifically for Indian-accented speech using self-collected data. A competitor can copy a live-call feature in a sprint. They cannot instantly replicate a fine-tuned model backed by a real, measured accuracy improvement on Indian speech — that requires actually doing the data collection and training work.

5. Product Vision — What Is Actually Being Built

VoxGuard is **not a single app**. It is one AI engine delivered through three different surfaces, each aimed at a different point in the adoption curve.

Surface	Form factor	Status	What it does
Clip Analysis Mode	Web app (browser, mic + file upload)	P0 — build & demo now	User uploads or records a clip; gets an instant real-vs-cloned verdict with a confidence score
Live Call Mode	Web app, simulated call via microphone	P1 — stretch goal	Simulated live call with a continuously updating on-screen risk meter
Enterprise Integration	Backend API / SDK, invisible to end user	P2 — roadmap, not built now	Bank/telecom embeds this directly into their own call-center software or network

5.1 Why a web app, not a native mobile app, for the prototype

- Browsers can access the microphone directly (the getUserMedia API) — no app-store install friction, and judges/testers can try it instantly on a laptop.
- It is dramatically faster to build and deploy in a hackathon timeframe than a native Android/iOS app.
- The **same backend API** that powers the web app is exactly what gets reused for the P2 enterprise integration — nothing built for P0 is wasted work.

5.2 How a citizen is actually protected, honestly stated

- **Self-check path (buildable now):** the citizen opens the app and manually checks a suspicious call they just recorded or received.
- **Passive protection path (P2, requires institutional adoption):** their bank or telecom has integrated VoxGuard on their end, so the citizen is protected without installing anything — but this only exists once an institution actually adopts the P2 integration. It should not be presented as already working for an individual today.

6. Complete Technical Architecture

6.1 Tech stack (exact)

Layer	Technology	Why
Core model	AASIST — pretrained checkpoint (clovaai/aasist, open-source; weights on HuggingFace)	Current strongest open baseline for anti-spoofing; see § 7 for full detail
ML framework	PyTorch	Matches the official AASIST implementation; avoids reimplementation risk
Feature extraction	None needed	AASIST is end-to-end on raw waveform — older approaches needed hand-crafted LFCC/MFCC features; AASIST removes that step entirely
Backend	Python + FastAPI	Fast to prototype, async-friendly for streaming
Real-time layer	FastAPI + WebSocket, ~4-second rolling audio chunks	A genuine real-time claim requires streaming analysis, not single-file upload
Database (P1)	PostgreSQL / Firebase	Call logs + risk scores for a dashboard — not core to the MVP
Demo deployment	Local GPU / laptop, not cloud	Avoids network-latency failure risk during a live demo — nothing should be able to break on stage
Containerization	Docker	Consistent environment across development and deployment

6.2 Alternatives considered and rejected

Alternative	Why not chosen
RawNet2	Open-source and viable, but an older, slightly weaker baseline than AASIST on standard benchmarks
RawGAT-ST	Same category as RawNet2 — AASIST is the stronger current baseline to build on
Hand-crafted features (LFCC/MFCC) + classical ML	Requires a separate feature-engineering step; AASIST’s end-to-end design removes this complexity entirely

6.3 The three-mode pipeline, step by step

Mode 1 — Clip Analysis (P0, the core demo)

- 1 User uploads a pre-recorded clip or records a new one directly in the browser.
- 2 The user (or the system, in test mode) selects which reference set to compare against — a standard ASVspoo sample, or a self-recorded Hindi/regional-accent clip.
- 3 The raw waveform is passed directly into AASIST — no separate feature-extraction stage.

- 4AASIST outputs a confidence score: real vs. cloned, with a percentage.
- 5The interface displays a **before/after comparison** — the model's accuracy before and after Indian-accent fine-tuning. This is the single most important demo moment: it is the concrete, measurable proof of the project's actual contribution.

Mode 2 — Live Call (P1, if time allows)

- 1A simulated call starts; the browser microphone begins capturing audio.
- 2Audio is streamed to the backend continuously via a WebSocket connection in ~4-second rolling chunks (mechanics explained fully in § 6.4).
- 3Each chunk is preprocessed (resampled, volume-normalized) and scored independently by AASIST.
- 4Scores are smoothed across chunks (e.g. rolling average, or “flag if 2 of the last 3 chunks are high-risk”) so one noisy chunk doesn't trigger a false alarm.
- 5A live on-screen meter updates continuously (green/yellow/red); results are optionally logged to a dashboard (PostgreSQL/Firebase).

Mode 3 — Enterprise Integration (P2, roadmap only — not built in the prototype)

- 1A bank or telecom's own system sends call audio to VoxGuard's hosted API/SDK.
- 2The same AASIST engine returns a risk score, hosted centrally.
- 3High-risk calls trigger an escalation recommendation (callback / multi-factor authentication).
- 4The alert is written into the blockchain-anchored audit ledger (§ 9).
- 5A human at the institution makes the final decision — the system never auto-blocks (§ 8).

6.4 The continuous chunking mechanism, in full detail

This directly answers a question that came up repeatedly during development: **does the system check once, or continuously?** The answer is continuously, for the entire duration of the call — not a single check at the start.

Step	What happens to each 4-second chunk
1. Capture	The WebSocket connection sends one 4-second audio clip as it's captured, live.
2. Preprocess	Resample to a standard rate and normalize volume.
3. Score	AASIST scores that specific chunk independently.
4. Update	The live meter updates using a score smoothed across recent chunks, not a single raw chunk.
5. Repeat	This entire cycle repeats roughly every 4 seconds until the call ends.

Why 4 seconds specifically: shorter windows (e.g. 1 second) don't give the model enough audio to reliably read pitch and prosody patterns, producing a jittery, unreliable score. Longer windows (e.g. 15 seconds) are more accurate but arrive too late to be useful mid-conversation. Roughly 4 seconds is the field-standard sweet spot for near-real-time synthetic-speech detection.

Background scoring vs. visible alerts — an important distinction: the background score refreshes silently every ~4 seconds for the whole call. The *visible* alert to the human, however, only fires the moment the risk level actually worsens (green to yellow, or yellow to red) — not on every single check while nothing has changed. Repeating an alert every 4 seconds while everything is fine would cause “alert fatigue,” the same reason a phone doesn't beep every time it **doesn't** find a virus. In a typical flagged call, the first visible warning might appear around the 12-second mark, with an escalated warning at 16 seconds — not only at the

very start or the very end of the call.

7. The Core AI Model — AASIST, in Full

7.1 What AASIST stands for and what it does

AASIST = **A**udio **A**nti-Spoofing using **I**ntegrated **S**pectro-Temporal graph attention networks. It is a deep learning model purpose-built to distinguish real human speech from synthetic or replayed speech.

7.2 Complete history

Detail	Information
Paper	“AASIST: Audio Anti-Spoofing Using Integrated Spectro-Temporal Graph Attention Networks”
Authors	Jee-weon Jung, Hee-Soo Heo, Hemlata Tak, Hye-jin Shim, Joon Son Chung, Bong-Jin Lee, Ha-Jin Yu, Nicholas Evans
Institutions	NAVER Corporation (South Korea), EURECOM (France), University of Seoul, KAIST
First published	arXiv, October 2021
Formal publication	ICASSP 2022 (IEEE Intl. Conference on Acoustics, Speech & Signal Processing), Singapore, May 2022
Core innovation	Earlier systems needed multiple separate sub-models, each tuned to a different type of fake-audio artifact (some show up in frequency, some in timing). AASIST unified this into a single model using a graph attention network that reads spectral and temporal clues together.
Result	Beat the previous best system by 20%. A smaller variant, AASIST-L, uses only ~85,000 parameters and still outperformed prior systems — tiny by AI standards, and cheap to run (relevant to § 14).
Field status	One of the most widely cited baseline models in anti-spoofing research since 2022 — a standard reference point, not an obscure choice.

7.3 The exact performance gap this project is built around

Pretrained AASIST scores 0.83% EER (*Equal Error Rate — lower is better*) **on the lab-quality ASVspoof2019-LA benchmark**. Published research shows that same model's error rate **degrades to roughly 43% EER** when tested on the “In-the-Wild” real-world dataset. That collapse in performance — lab-clean audio vs. real, messy, compressed audio — is the actual research gap this project targets, specifically for Indian-accented speech.

This number is the single strongest, most defensible claim in the entire project: it is not a marketing statement, it is a documented property of the field's own standard benchmark, and this project's contribution is measuring and narrowing that gap for a specific, underserved population (Indian languages and accents) using real, collected data — detailed fully in § 8.

7A. What Is ASVspoof?

ASVspoof (Automatic Speaker Verification Spoofing and Countermeasures Challenge) is the standardized public benchmark the entire anti-spoofing research field uses to prove a detection model actually works — the same role a public, agreed-upon exam plays versus every student just claiming their own score.

Year	What changed
2013	A special research session, mainly to raise awareness that this was a real problem worth studying.
2015	First real competitive challenge — detecting text-to-speech and voice-conversion fakes.
2017	Focus shifted to “replay attacks” (playing a recording of someone’s real voice back through a speaker).
2019	Combined all three attack types into one challenge. This is the most widely used version, and the one AASIST’s 0.83% EER figure is measured against.
2021	Added more realistic, compressed/degraded audio conditions, closer to real phone-call quality — this is the version where detection scores generally dropped.
2024 (ASVspoof5)	Crowdsourced data at much larger scale, and added adversarial attacks specifically designed to fool detectors.

It is organized by a rotating academic consortium — the University of Edinburgh, EURECOM, the University of Eastern Finland, and Japan’s National Institute of Informatics have been consistently involved, led by researchers including Junichi Yamagishi and Nicholas Evans (the same Nicholas Evans who co-authored the AASIST paper). Each edition releases a free public dataset of real and fake speech plus a standard scoring method, so results from different research teams are directly comparable.

8. The Actual Research Contribution — Closing the Indian-Accent Gap

This section is the most important one for an implementation team to execute correctly, because it is the part of the project that is genuinely novel work rather than assembling existing components.

8.1 The approach: fine-tune, don't retrain

Do **not** train AASIST from scratch — that is unnecessary, expensive, and throws away a model already validated on a huge public benchmark. Instead:

- 1 Start from the pretrained AASIST checkpoint (clovaaai/aasist, weights available on HuggingFace).
- 2 **Freeze** the early layers (which already learned general spectro-temporal spoof artifacts) and **fine-tune only the final layers** on a small, purpose-built Indian-speech dataset.
- 3 This is a light, fast training job — hours, not days — because most of the model's learned knowledge is being reused, not rebuilt.

8.2 Using the team's own Indian-language / dialect dataset

The team has stated it already has a dataset of Indian-language and dialect speech to use for this fine-tuning step. The following is the concrete, actionable plan an implementation team should follow to turn that raw dataset into a working fine-tuned model:

Step 1 — Audit and label the dataset

- Confirm every sample is labeled unambiguously as **real** (a genuine human recording) or **cloned/synthetic**. If the dataset only contains real speech, the synthetic/cloned class must be generated (see Step 2) — a model cannot learn to detect fakes without seeing examples of fakes.
- Record metadata per sample: language, dialect/accent, speaker ID (if available), recording condition (clean studio vs. phone-call-quality/compressed audio), and sample duration.
- Check for class imbalance — anti-spoofing datasets are often heavily skewed toward real speech; if so, either collect/generate more synthetic samples or apply class-weighting during fine-tuning.

Step 2 — Generate the synthetic/cloned class, if needed

- Use open, freely available voice-cloning tools — **Coqui TTS** or **RVC (Retrieval-based Voice Conversion)** — to generate cloned versions of the team's own or consenting teammates' voices.
- **Ethical requirement, not optional:** every cloned sample used for training or demo must be of the team's own voice or a consenting teammate's voice, clearly disclosed. This is both an ethical necessity and a credibility point if challenged.
- Generate cloned samples across the same languages/dialects/speakers present in the real-speech side of the dataset, so the model learns to compare like with like.

Step 3 — Split the dataset properly

Split	Purpose	Suggested proportion
Train	Used to actually update the model's final-layer weights	~70%
Validation	Used during training to tune hyperparameters and catch overfitting early	~15%
Test (held-out)	Never touched during training — used only for the final, honestly-reported accuracy number	~15%

Critically, split by **speaker**, not just by individual audio clip — if the same speaker appears in both train and test sets, the reported accuracy will be artificially inflated because the model may partly be recognizing the specific speaker rather than genuinely detecting spoofing.

Step 4 — Fine-tune

- Fine-tune only the final classification layers of the pretrained AASIST checkpoint on the training split, using standard cross-entropy loss for the real-vs-cloned binary classification task.
- Monitor validation EER during training; stop when validation performance plateaus (early stopping) to avoid overfitting to the small fine-tuning dataset.

Step 5 — Evaluate honestly, and report the before/after

- Measure EER on the held-out test split **before** fine-tuning (using the original pretrained checkpoint) and **after** fine-tuning.
- Report the result **per language/dialect** where the sample size allows — this per-language transparency is the specific claim competitors don't currently make publicly.
- Report the honest number even if it's not flattering. A credible, modest improvement (e.g. 43% EER reduced to a meaningfully lower number on Indian-accented data) is more valuable to the project's credibility than an inflated or unverifiable claim.

This before/after comparison — built directly from the team's own dataset — is the single strongest, most concrete demo asset in the entire project. It should be the centerpiece of any live demonstration, not a supporting detail.

9. Trust, Safety & Human-in-the-Loop Design

No AI voice detector — this one or any competitor's — is ever fully accurate. Trust is not achieved by claiming the AI is always right; it is achieved by designing the system so that a wrong answer never directly causes harm.

9.1 The core design rule

*The AI **never takes the final action itself**. It never hangs up a call, freezes an account, or auto-approves/auto-blocks a transfer. It only informs a human decision. A human (or a simple deterministic rule) is always the last checkpoint before anything consequential happens.*

9.2 Walkthrough: how a real flagged call is handled

- 1 A call starts; VoxGuard begins scoring in the background without interrupting anything.
- 2 The risk score crosses into a high-risk zone (e.g. 82/100).
- 3 The system displays a warning, not a block: “82% chance this voice is synthetic — recommend verifying via callback to the registered number before proceeding.”
- 4 A hard rule can force extra caution independent of the AI's score for high-stakes situations — e.g. “any transfer above Rs. 1 lakh always requires callback verification, regardless of score.”
- 5 The human decides — calls back the verified number, confirms the request is fake, and stops the fraud.
- 6 The outcome (proceeded / verified / escalated) is logged either way for later review.

9.3 Handling both failure modes

Failure type	What happens	Mitigation
False negative	The AI misses a real clone — scores it as low-risk	High-value transactions get mandatory callback verification regardless of AI score; continuous retraining as cloning tools evolve
False positive	The AI flags a real, genuine voice as suspicious	The system never auto-blocks — worst case is a moment of friction (a verification step), never a blocked call

Worst case in either failure mode: a delay, not a loss. That is the actual, honest trust argument for this system — not that the AI is infallible, but that its mistakes are designed to be cheap.

10. Privacy, Data Storage & Legal Compliance

10.1 What gets stored, tiered by risk

The system does **not** store full audio/conversation content by default for every call, regardless of its flag color. A tiered policy applies:

Flag	What is stored	Retention
Green (safe)	Call ID + final score only — no audio	Auto-deleted within 24–48 hours
Yellow (medium risk)	Risk score + timestamp, no audio	~30–90 days (useful for later threshold tuning)
Red (high risk)	Risk score, an extracted numerical “voice fingerprint,” timestamp, call metadata, and the human’s eventual decision	Only as long as legally necessary for a possible investigation

10.2 The “voice fingerprint,” not the recording

Even for high-risk (red) calls, the system does not need to store the actual recording. It extracts a mathematical fingerprint of the voice — numbers describing pitch, spectral shape, and similar properties — which is what AASIST actually needs to make its decision. This fingerprint cannot be played back or understood as speech. This is the same principle apps like Shazam use: they don’t store the song you hummed, only a fingerprint of it.

10.3 User awareness and consent

- A spoken notice at call start, extending the now-universal “this call may be recorded for quality purposes” disclosure: *“This call is monitored by an AI fraud-detection system. Only a risk score is stored, not the recording.”*
- A one-time consent step when a user first signs up for the service through a bank or institution’s app.

10.4 Legal grounding — India’s DPDP Act, 2023

This isn’t just good practice — it is now Indian law. The **Digital Personal Data Protection (DPDP) Act, 2023** had its Rules formally notified on **13 November 2025**, with phased enforcement through **13 May 2027**, and penalties of up to **Rs. 250 crore** for violations. Relevant requirements this system’s design directly satisfies:

DPDP requirement	How VoxGuard satisfies it
Clear notice & stated purpose	Explicit spoken/written disclosure before monitoring begins
Data minimization	Voice fingerprints and scores are stored, not raw recordings, by default
Storage limitation	Tiered auto-deletion policy (§ 10.1)
Right to erasure	Users can request deletion of stored records
Breach notification (72 hrs)	Incident response process must be defined as part of production deployment

10.5 How deletion actually works technically

Because raw audio isn't stored by default, the total footprint is small — a handful of numbers per call, not hours of audio. A scheduled background job (the same mechanism an email inbox uses to clear spam after 30 days) periodically scans the database and deletes anything past its retention window. The blockchain ledger (§ 11) does not bloat from this, because it only ever stores tiny cryptographic hashes, never audio.

11. The Blockchain Audit Ledger

11.1 What it is

A tamper-evident logbook: once an entry is written, nobody — including the team running the system — can secretly edit or delete it without the change being detectable. Each new log entry includes a cryptographic fingerprint (“hash”) of the previous entry; changing an old entry breaks every fingerprint after it, and that break is instantly detectable. This is the core mechanism behind blockchain, without any of the cryptocurrency machinery.

11.2 Why it's worth doing

- **Evidence for investigations** — proves exactly what was flagged and when, if a bank later disputes whether the system actually raised a warning.
- **Protects against insider tampering** — an employee involved in a fraud cannot quietly delete the evidence that the system warned them.
- **Regulatory trust** — banks and RBI-regulated institutions require auditable records.
- **Directly ties to the Problem Statement's theme** (“Blockchain & Cybersecurity”) with a real, functional justification — not decoration.

11.3 Honest scoping note

A full public blockchain (like Bitcoin) is unnecessary here. A lightweight private hash-chain achieves the same tamper-evidence at a fraction of the cost and complexity — worth stating explicitly if asked “why not just use a database?”

12. Threat Model — Defending Against Spoofed Known-Contact Calls

12.1 The scenario

A scammer spoofs the caller ID of someone already saved in the victim's contacts (e.g. “Dad” or “Boss”) **and** clones that person's voice. This is not an edge case — it is the central scam pattern this entire project exists to catch. A naive design that skips analysis for “trusted, saved contacts” (for privacy or performance reasons) would create a blind spot exactly where the attacker is aiming.

12.2 Why caller ID cannot be the filter

Caller ID spoofing is a documented telecom-network weakness — legacy signaling (SS7) does not cryptographically verify that a displayed number belongs to the actual caller. “This call says it’s from Dad” and “this call is actually from Dad” are two unrelated facts.

12.3 The actual solution: analyze everyone, add a second check for known contacts

Caller type	Checks applied
Unknown number	Standard AASIST anti-spoofing check
Matches a saved contact	AASIST anti-spoofing check plus a personal voiceprint match against that specific contact's known voice

Anti-spoofing (AASIST) answers “is this a real human voice, or synthetic?” **Speaker verification** (voiceprint matching) answers a different question: “does this voice match the specific person it claims to be?” For a saved contact, with the user's permission, a lightweight voiceprint can be built from a known-genuine sample of that person's voice; new calls claiming to be that contact are checked against both models. If either check fails, a high-risk alert fires regardless of what the caller ID displays.

This connects directly back to the underlying research field: ASVspoof's full name is “Automatic **Speaker Verification** Spoofing” countermeasures — the field was built around exactly this combined problem (verifying claimed identity *and* detecting spoofing), not spoof detection alone. Adding voiceprint matching for known contacts is a return to that original framing, not a bolt-on feature.

12.4 User configuration, with an informed default

- **Default:** analyze all calls, known and unknown — because known contacts are the highest-value target for this specific attack.
- If a privacy-conscious user disables analysis for saved contacts, show an explicit warning first (the same pattern antivirus software uses before disabling real-time protection), rather than silently honoring the toggle.

12.5 Honest limitation

Caller ID spoofing itself is a telecom-network problem, not something a phone-side app can fully fix — that requires network-level protocols such as STIR/SHAKEN (used in the US/Canada) or equivalent efforts by India's telecom regulator (TRAI). This system's honest position: it cannot stop a number from being spoofed, but it never trusts the number alone — it listens to the actual voice regardless of what the caller ID says.

13. Cost Justification — Is Commercial AI Too Expensive Here?

This is a reasonable concern, but it is based on the cost profile of **generative AI** (large language models with billions of parameters). AASIST is a fundamentally different, much cheaper category of model.

Consideration	Reality
Model size	AASIST-L has only ~85,000 parameters — closer to a small classical ML model than an LLM
Training cost	Only the final layers are fine-tuned on a small, self-collected dataset — hours of compute, not weeks
Inference cost per call	Classifying a 4-second clip is a lightweight computation; can run on a normal laptop GPU or even CPU
Cost of doing nothing	India is projected to lose ~Rs. 70,000 crore to deepfake fraud in 2025 — a fraction-of-a-rupee detection cost per call is trivial by comparison
Business model	Priced the same way institutions already pay for SMS OTPs or transaction monitoring — a small incremental cost, not a wholesale AI investment
Future cost reduction	The P2 on-device/edge deployment option lets an institution run inference on its own hardware, avoiding per-call third-party API fees entirely

One-line answer for a skeptical judge or investor: “We’re not deploying a large generative model — we’re running a lightweight, 85K-parameter classifier costing fractions of a rupee per call, against a fraud problem costing India Rs. 70,000 crore a year.”

14. Feasibility & Viability

	Key points
FEASIBILITY	Public benchmarks (ASVspoof, WaveFake) mean no restricted stakeholder data is needed. Self-recorded, ethically-consented Indian-accent data is fully within the team's control. Supervised audio classification via AASIST is a proven, deployable architecture, not unproven research.
VIABILITY	Fake-CEO / family-emergency voice fraud is a global, India-reported, escalating threat (§ 2.3). Buyers — banks, NBFCs, enterprises, telecom, public institutions — are AICTE-aligned. Three funded Indian competitors validate that real market demand exists.
CHALLENGES & RISKS	The gap between clean benchmark data and real, compressed telephony/VoIP audio (§ 7.3). Cloning tools keep improving — an ongoing adversarial arms race. Voice biometric data requires privacy-first handling from day one (§ 10).
STRATEGIES	Tunable risk thresholds with mandatory human-in-the-loop escalation, never full auto-block (§ 9). Own recorded Indian-accented dataset directly closes the accent/language gap (§ 8). Pilot-first rollout with a single bank/telecom partner before wider integration.

15. Measurable Impact & Success Metrics

< 1 sec	0 of 3	Zero
Target real-time risk-score latency per call	Funded Indian competitors currently publishing per-language accuracy	Bank / government data needed to build & validate the model

Capability	DetectifAI	Neural Defend	Kroop AI	VoxGuard
Real-time in-call risk score	Yes	Partial	Partial	Yes
Reaches citizens / smaller institutions	Early-stage	Yes (via Zee News)	No (enterprise-only)	Yes, by design
Per-Indian-language accuracy published	Unverified	Unverified	Unverified	Yes — planned as a core commitment
Blockchain-anchored tamper-evident log	Unverified	Unverified	Unverified	Yes
Public-sector / education deployable	Unverified	Unverified	Unverified	Yes, by design

“Unverified” reflects that this project team has not confirmed these details against competitors’ primary technical documentation — see § 4.1. Do not present these as confirmed facts to a technical audience without independent verification.

Core measurable target: an Equal Error Rate benchmarked against the standard ASVspoof protocol (not an invented number), reported both overall and per Indian language/dialect where sample size allows, before and after fine-tuning (§ 8.2, Step 5).

16. Risks & Open Challenges — Stated Honestly

Category	Risk
Technical	Generalization from benchmark data to real, noisy, compressed telephony audio; cloning tools will keep evolving to evade detection (an ongoing arms race, not a one-time fix)
Data	Indian-accented cloned-voice data does not exist publicly at scale — must be collected and generated ethically by the team, which takes real time to do properly
Ethics / privacy	Handling voice biometric data requires a clear minimal-retention, privacy-preserving design from the start, not bolted on afterward (§ 10)
Adoption	Integration into real banking/telecom infrastructure requires partnerships and security certification that take far longer than a hackathon timeframe — the roadmap should be framed honestly as pilot-first (§ 6.3, Mode 3)
Competitive	At least one competitor (DetectifAI) already offers live-call audio detection — the “real-time” claim alone is no longer uniquely differentiating (§ 4)
Network-layer	Caller ID spoofing itself cannot be fixed by this system alone — it is a telecom-layer problem (§ 12.5)

17. Build Roadmap

Phase	Scope	Must include
P0 — Core prototype	Clip Analysis Mode as a web app	Upload/record, then classify with confidence; tested on both ASVspooof samples AND self-recorded Hindi/regional data; the before/after fine-tuning comparison (§ 8.2)
P1 — Stretch goal	Live Call Mode simulation	Real-time streaming risk meter during a simulated call; a simple call-log dashboard
P2 — Roadmap (post-hackathon)	Enterprise integration	Robustness across 3+ Indian languages; integration-ready API/SDK for banks/telecom; on-device/edge inference; blockchain audit ledger (§ 11); personal voiceprint matching for known contacts (§ 12)

18. Research References

Reference	Link / source
AASIST paper (Jung et al., ICASSP 2022)	arxiv.org/abs/2110.01200
AASIST official code	github.com/clovaai/aasist
ASVspoof 2021 (Yamagishi/Liu et al.)	arxiv.org/abs/2210.02437
ASVspoof 2021 evaluation plan companion paper	arxiv.org/abs/2109.00535
Audio Deepfake Detection: A Survey (Yi, Wang, Tao et al., 2023)	arxiv.org/abs/2308.14970
India deepfake fraud loss projections	Pi-Labs “Digital Deception Epidemic” report
Deepfake scam victimization survey	McAfee consumer survey (India)
DPDP Act, 2023 & Rules (notified 13 Nov 2025)	Ministry of Electronics & IT (MeitY) official notification
Competitor research	Public funding/press coverage of DetectifAI, Neural Defend, Kroop AI — not verified technical documentation (§ 4.1)

19. Project Naming — Status

The working name used throughout this document is **VoxGuard**. During development, the team also explored alternatives without reaching a final decision:

Category	Options discussed
Direct / literal	VoxGuard, CallShield, EchoGuard
Truth / authenticity angle	VeriVox, TrueVox, AuthentiCall
Professional Sanskrit-root	Setu, Vishwas, Satya
Hindi / citizen-facing	AsliAwaaz, Pehchaan, VaaniRaksha

Note: avoid “Kavach” — it is already the real name of Indian Railways' anti-collision safety system and risks confusion with an existing government brand. If the team finalizes a different name, it should be substituted consistently across this document and all product surfaces before external distribution.

Glossary of key terms

Term	Plain-language meaning
EER (Equal Error Rate)	A single accuracy number for a detector — lower is better. It's the point where the rate of missed fakes equals the rate of false alarms.
Spoofing / anti-spoofing	“Spoofing” = faking a voice; “anti-spoofing” = detecting that a voice is faked.
Speaker verification	Checking whether a voice matches one specific claimed person, as opposed to just checking whether a voice is real at all.
Voice fingerprint / embedding	A set of numbers describing a voice's characteristics, extracted by a model — not audio, and not something that can be played back.
WebSocket	A persistent, two-way internet connection that lets audio (or any data) stream continuously, instead of the client having to ask again and again.
Hash / hash-chain	A hash is a short digital fingerprint of a piece of data. Chaining hashes together means changing any old entry breaks every fingerprint after it — the basis of blockchain-style tamper evidence.
Fine-tuning	Taking an already-trained model and further training only part of it on a new, smaller, specific dataset — much cheaper than training from scratch.
P0 / P1 / P2	Priority tiers: P0 = must work for the live demo, P1 = stretch goal if time allows, P2 = roadmap beyond the hackathon.